

Titanic Data Cleaning and Preprocessing Report

Introduction

Data preprocessing is one of the most important steps in any data analysis or machine learning project. Real-world datasets often contain missing values, inconsistent formats, and unnecessary information that can affect the quality of analysis and model performance.

In this project, the Titanic passenger dataset is cleaned and prepared for further analysis. The dataset contains information about passengers aboard the Titanic, including demographic details, ticket information, and survival status. The main objective is to identify data quality issues, handle missing values, standardize the dataset structure, and create a clean version that can be used for exploratory data analysis (EDA) and machine learning tasks.

The final cleaned dataset is saved as `titanic_cleaned.csv`.

Dataset Overview

The original dataset contains 891 passenger records and 12 features. Each row represents a passenger, while each column contains information such as age, gender, passenger class, ticket details, and survival status.

Column Name	Description
PassengerId	Unique passenger identifier
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class
Name	Passenger name
Sex	Passenger gender
Age	Passenger age
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Ticket	Ticket number
Fare	Ticket fare
Cabin	Cabin number
Embarked	Port of embarkation

Data Cleaning Process

1. Initial Data Inspection

The dataset is first loaded using the Pandas library and inspected to understand its structure, data types, and missing values.

The following missing values are identified:

- Age: 177 missing values
- Cabin: 687 missing values
- Embarked: 2 missing values

This step helps determine the appropriate cleaning strategy for each feature.

2. Handling Missing Values

Different techniques are used depending on the nature of the column.

Age

Since age is a numerical feature, missing values are replaced using the median age of the dataset (28 years). The median is chosen because it is less sensitive to extreme values than the mean.

Cabin

A large portion of cabin information is missing. Instead of removing the column, missing entries are replaced with the label "Unknown" so that information about missing cabins is preserved.

Embarked

Only two values are missing in this column. These entries are filled using the most frequently occurring embarkation port, S (Southampton).

3. Duplicate Check

The dataset is examined for duplicate records using Pandas' duplicate detection functions.

No duplicate rows are found in the dataset, so no records need to be removed.

4. Data Type Conversion

To improve consistency and memory efficiency, several columns are converted to more suitable data types.

The following columns are converted to the category data type:

- Survived
- Pclass
- Sex
- Embarked

The Age column is converted to an integer format after missing values are handled.

5. Column Renaming

To make the dataset easier to read and work with, all column names are renamed using the snake_case naming convention.

Original Name	New Name
PassengerId	passenger_id
Survived	is_survived
Pclass	passenger_class
Name	passenger_name
Sex	gender
Age	age
SibSp	siblings_spouses_count
Parch	parents_children_count
Ticket	ticket_number
Fare	fare_paid
Cabin	cabin_code
Embarked	embarkation_port

Using consistent naming conventions improves readability and follows common data science practices.

Final Dataset

After preprocessing, the dataset contains:

- No missing values in Age
- No missing values in Embarked
- Missing Cabin entries replaced with "Unknown"
- Standardized column names
- Optimized data types for categorical features
- No duplicate records

The cleaned dataset is exported as `titanic_cleaned.csv` for future analysis and machine learning applications.

Conclusion

This project focuses on preparing the Titanic dataset for further analytical tasks. Missing values are handled, data types are optimized, column names are standardized, and the overall dataset structure is improved.

The resulting dataset is cleaner, more consistent, and better suited for exploratory analysis, visualization, and predictive modeling. Proper preprocessing ensures that future analyses are based on reliable and well-structured data, making it an essential step in the data science workflow.